



Deep Rooting as an Indicator of Deep Soil Water and N Uptake in Potato (*Solanum tuberosum* L.)

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Abstract

Breeding for potato deep roots can increase water and nitrogen uptake by potatoes and it can be an option to maintain stable yields with decreased inputs. This study investigates the relationship between potato root characteristics, water stress resistance and deep soil nitrogen uptake, accessing variations among cultivars and nitrogen fertilization levels. Thirteen potato cultivars were grown during 2018 and 2020 at a semi-field root phenotyping platform in Denmark. Root growth was monitored via minirhizotron tubes down to 1.8 m soil depth. Drought treatment started in the mid-June and deep soil nitrogen uptake was tracked via ^{15}N isotope application at 1.3–1.4 m soil depth during tuber formation. Water stress resilience was identified using ^{13}C natural discrimination process in plants. Tuber samples were analyzed for ^{15}N and ^{13}C content. While drought affected potato yield (not always significantly), it did not affect nitrogen uptake. Root length and distribution varied among varieties, with deeper roots (down to 1.30 m) observed in August. Statistical differences ($p < 0.05$) in root length, yield and nitrogen uptake were found among varieties. Cultivars with longer growing season exhibited larger, deeper roots and increased nitrogen uptake from deep soil. High correlation ($R = 0.8$) between deep roots and ^{15}N uptake was observed for all varieties. Deeper roots are contributing to deep soil nitrogen uptake, but ^{13}C content in tubers is not a reliable indicator of water stress resilience. Despite this, the study suggests the potential for breeding potatoes with deep roots to achieve stable yields, considering differences in water and nitrogen uptake among varieties.

Keywords Deep roots · Isotope tracers (^{13}C and ^{15}N) · Nitrogen uptake · RadiMax root phenotyping · *Solanum tuberosum* L. · Water stress

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Introduction

Potatoes (*Solanum tuberosum*) are mainly grown in Asia and Europe (80% of total potato production) with China being the largest potato producer in the world (FAOSTAT 2022). With its total world production of approximately 350 million tons per year, it qualifies as one of the major staple crops in the global food supply (FAOSTAT 2022). Denmark cultivates potatoes across an expanse of approx. 56 000 ha, yielding an average of 40 t ha⁻¹. The significance of potato production in Denmark lies in its crucial role in human nutrition, as nearly all cultivated potatoes are intended for human consumption (Statistik 2022). Potato production is challenged by many factors: drought sensitivity and low nitrogen use efficiency of potatoes, high vulnerability for many diseases and pests, high costs of seed material production and management etc. Moreover, having in mind that potatoes are mostly grown in sandy soils (Ahmadi et al. 2011) and due to its rather sparse and shallow root system (Iwama 2008; Stalham and Allen 2001), water stress and loss of nutrients are almost inevitable in potato production. As climate change scenarios are predicting more drought spells and as increased nitrogen use efficiency (NUE) of cropping systems is crucial for sustainable intensification, it is obvious that these challenges to potato production are only going to increase in the future. Therefore, research and development is necessary in order to respond to the challenges and to prepare potato production for green transition.

Potato yields are substantially reduced by water stress and previous research showed that high drought sensitivity of potatoes is mostly due to their shallow root system and low capacity for recovery after a period of water stress (Iwama and Yamaguchi 2006). Several studies (Albiski et al. 2012; Boguszewska-Mańkowska et al. 2020; Lahlou and Ledent 2005) have examined the effects of drought on root growth (total root length over measured area and maximum root depth), but these studies find seemingly contradictory results. These contradictions are most probably a result of complex G×E interactions, experimental differences (duration, severity and timing of drought, soil type and compaction etc.) and lack of adequate and accurate techniques for quantifying root growth (Boguszewska-Mańkowska et al. 2020; Iwama 2008; Joshi et al. 2016). Additionally, potato has low (33–56%) nitrogen uptake efficiency (NUE) (Errebhi et al. 1998), when compared to NUE (57%) of wheat (Ladha et al. 2005). Low potatoes' NUE is usually due to its limited root zone, short growing season (Zebarth et al. 2004), inefficient reallocation of shoot nitrogen into tubers and loss of nitrogen through leaching. Previous studies (Errebhi et al. 1999; Getahun et al. 2020) showed that the nitrogen demand and uptake varies by variety and is related to time to maturity, root density and root depth.

Breeding for increased water and nitrogen uptake by potatoes is an option to maintain stable yields with decreased inputs. Yet, it is a complex task, as genotypic variations in nitrogen uptake and response to drought stress cannot be attributed to a single plant characteristic due to performance variations across years and locations (G×E). Research primarily focuses on above-ground traits and potatoes' tuber characteristics to mitigate water stress and increase nitrogen

uptake. Breeding programmes for drought tolerance and high NUE do not include selection for root performance, despite established links between root growth, drought tolerance and N uptake variations among varieties (Lahlou and Ledent 2005). Successful “root breeding” strategies depend on root phenotyping for identification of heritable root traits important for increased crop efficiency in nutrient and water uptake. Identification and validation (linking the traits to improved root function) of such root traits are often limited by the root phenotyping method. Ideally, the method should have the ability to accurately and efficiently phenotype roots of many genotypes to the full crop maturity and under field conditions. Moreover, such a technique/method should be able to account for the deep and thin root structures, which are easily lost when destructive root measuring techniques are used, and it should be able to account for G × E interactions. Additionally, in order to create root data for breeding, it is important that the technique allows observation of varietal differences in the root traits among many genotypes under realistic growth conditions.

Potato roots have been quantified under both field and greenhouse/pot conditions. Literature offers many different methods for measuring potato roots in the field—core sampling (Iwama 2008) or excavation by a combination of a large and small hand forks to first loosen the root and then to extract roots from soil (Wishart et al. 2013). Most of these methods are labour and time demanding, recovery of fine thin roots (less than 2 mm thick) is depending on the time invested and the person performing the measurements. However, in field trials, many genotypes can be screened under realistic conditions (environment and management practices) and plants can be grown to full maturity thus allowing growth of deep roots later in growing season. In lab-scale studies, potato plants can be grown in aeroponics (Rykaczewska et al. 2018), in bioassay plates, 0.24 × 0.22 m rhizotrons for 10 days (Kirchgesser et al. 2023), etc. Some of these methods allow recovering fine roots to a high degree, but the short experimental time and setup are missing out on the full and deep root development of a potato crop. In greenhouse experiments, pots can be specifically prepared to allow more realistic conditions for root development, like in the 1-m deep pots employed by Boguszevska-Mańkowska et al. (2020), which could be opened, enabling total root system extraction at the end of the experiment. Several attempts to select deep rooting genotypes have been made and different root traits have been taken into account, e.g., pulling resistance of roots (Ekanayake and Midmore 1992; Stalham and Allen 2004), ability of roots to continue to elongate under low water potential (Westgate and Boyer 1985)) and early vigorous root proliferation (Puértolas et al. 2014).

In order to access the root function, root traits shall be correlated to uptake activity. The uptake of nitrogen and water from deep soil layer is affected by root system-based limitations, such as the amount of plant available resources, sub-soil constrains and crop managing factors (Thorup-Kristensen and Kirkegaard 2016). In the RadiMax facility, soil is repacked and carefully managed with only light machinery; thus, most of sub-soil constrains are avoided. The presence of roots in deep soil layers does not show how much these roots are contributing to N and water uptake, making it important to link root data to estimates of the deep root activity. In this study, the stable nitrogen isotope (^{15}N) has been used as a tracer method for

deep soil nitrogen uptake and measurements of carbon isotope (^{13}C) in tubers have been used as proxy for estimating water stress resilience. Carbon isotope (^{13}C) discrimination is a natural process that occurs in plants during photosynthesis and it is affected by water stress (Gaudillère et al. 2002). Plants keep their stomata more closed when stressed by water limitation which is leading to reduced discrimination against ^{13}C and thereby less negative $\delta^{13}\text{C}$ values in plant dry matter.

We hypothesized that deep potato root growth (approx. 0.7–1.3 m soil depth) varies among genotypes and that this leads to significant differences in nitrogen and water uptake from deep soil layers and that deep rooting leads to improved water stress resilience in potatoes.

The specific hypotheses of this study were as follows:

1. Minirhizotron (MR) root phenotyping in combination with an AI-based software tool for root image analysis can be used for detection of genotypic variability in deep rooting of potatoes (> 0.7 m soil depth).
2. Potato cultivars with deep rooting will show higher uptake of ^{15}N applied to subsoil.
3. Potato cultivars showing deep rooting will show higher drought resilience as measured by ^{13}C discrimination.

Materials and Methods

Root Phenotyping Facility—RadiMax

The experiments were performed in 2018 and 2020 in parts of the RadiMax semi-field root phenotyping facility at the University of Copenhagen (Svane et al. 2019a). Each bed of the RadiMax facility is 40 m long and 9.7 m wide with a V-shaped bottom and equipped with minirhizotron tubes for root observation (Svane et al. 2019a). The V-shaped bottom is sealed with a membrane and has a 15.8° slope ranging from 0.4 m soil depth at the edges to 2 m in the centre.

Water Sensors

Each plot has 36 water sensors (time domain transmission sensors, Acclima Inc., USA), 8 middle sensors placed at two positions (East and West) in the centre of the bed at 0.5-m depth increments and 26 slope sensors placed 0.25 m above the sub-irrigation driplines at soil depths of 0.74, 0.87, 1.01, 1.15, 1.29, 1.42, 1.56, 1.70 and 1.83 m. The sensors provided readings of volumetric water content (VCW) and soil temperature every 5 min. The middle sensors' readings were used for evaluating the water content in the bulk soil of the bed. The slope sensors were placed close to the bottom of the bed, which affected the soil water dynamics strongly compared to the measurements in the bulk soil. The slope sensors above the driplines were used to monitor VCW for the isotope injection depth in the period around injection.

Minirhizotron Imaging System

Each of the RadiMax beds has 150 minirhizotron PMMA plastic tubes (5.5 long and 0.07 m outer diameter) placed at 25 cm distance. In this experiment, potato rows were planted at 0.5-m row distance (between two neighbour minirhizotron tubes), thus allowing us to use two minirhizotrons for root measurements of each row.

Sub-irrigation System

The RadiMax facility is equipped with pressure-compensated drip-irrigation lines that are placed along the length of the beds. Lines are parallel to each other placed at 0.2-m depth intervals, starting at 0.74 m soil depth. Each of the lines can be activated individually to irrigate certain soil depth at selected time intervals.

In 2018, the sub-irrigation system was used to create a water supply gradient with subsoil irrigation at increasing depth from the side towards the middle of the bed. This led to high water availability at the edge of the bed (sub-irrigated sample) and drier soil towards the middle of the bed (non-sub-irrigated sample), where the sub-irrigation was too deep for the roots to reach. The most shallow sub-irrigation line was activated when the water sensor at the corresponding depth showed onset of water use (middle of June 2018), and by the end of July, all lines were opened and the full soil profile was irrigated, as all water sensors showed water uptake.

In 2020, the sub-irrigation system was not used for irrigation, thereby creating an opposite water gradient, where water stress was based on soil volume available to the plants—deeper parts of the bed in the middle had a larger available soil volume (deep root sample) than the more shallow parts closer to the edges of the bed (shallow root sample).

Rainout Shelters and Climate Data

In order to create water stress conditions, the experiments were covered by movable rainout shelters from mid-July each year, thus preventing any precipitations during the remaining part of the growing season. The drought treatment started when potato crops were at the beginning of bulking phase, and water stress conditions in the soil were followed through daily reading of soil water and temperature sensors.

Meteorological measurements (reference evapotranspiration, air temperature and precipitation) were collected from a local weather station placed ca 600 m west of the facility.

Potato Cultivars and Planting Design

Selected cultivars of potato (*Solanum tuberosum* L.) were grown in 2018 and 2020 in a randomized complete block design. In 2018, seven cultivars were grown in two blocks in each RadiMax bed, while in 2020, six cultivars were grown at two N fertilizer levels in eight blocks, but all located in the same RadiMax bed. In both years,

each cultivar was grown in eight replicates, but with root measurements for four of the replicates as only one side of the plot was grown above minirhizotron tube allowing root imaging.

Varietal selection was based on the phenology, to have a uniform duration of growth. Most of genotypes were commercially established varieties; all were tetraploid *Solanum tuberosum*, mainly from the middle (100–120 days) maturity group, but with different market use. Two potato cultivars (Desiree and Aventura) were grown in both years.

Potato tubers of each cultivar were planted in single-row plots perpendicular to the centre aisle of the experimental beds, with 0.3 m within-row distance. The plots were positioned between two minirhizotron tubes with a 0.5-m row width (approx. 65,000 plants ha⁻¹). Major activities and details during two potato growing seasons are presented in Table 1.

Nitrogen Fertilization

Nitrogen fertilizer was applied in both years prior to planting at an average recommended dose for Denmark (140 kg N ha⁻¹). In 2020, extra 70 kg N ha⁻¹ (high

Table 1 Overview of the activities conducted in 2018 and 2020

Activity	2018	2020
RadiMax bed	3 and 4	3
Previous crop	2017: Perennial ryegrass 2016: Potato at bed 4	2019: Perennial ryegrass 2018: Potato
Sub irrigation	On from mid-June until the harvest	Not in use
Soil preparation (plowing + tillage)	14/04/18 19/04/18	26/09/19 14/04/20
Cultivars (root replicates, tuber replicates)	7 (<i>n</i> = 4 roots) (<i>n</i> = 8 tubers)	6 (<i>n</i> = 4 roots) (<i>n</i> = 8 tubers)
Fertilization (kg N ha ⁻¹)	140 prior to planting	140 prior to planting 70 given only to high N blocks at 07/07/2020
Planting	25–26/04/18	21/04/20
Planting density	65,000 (plants ha ⁻¹)	65,000 (plants ha ⁻¹)
Imaging dates	14–19/06/18 16–23/07/18 14–16/08/18 03–07/09/18	29/05/20 19/06/20 18/08/20
Isotope injection (date and depth)	20/07/18 140 cm	15/07/20 130–140 cm
Drought start	02/07/18	10/07/20
Harvest	27–28/08/18	25/08/20

nitrogen dose) was applied manually between the rows to the four blocks in the beginning of July, when potato plants were in the tuber bulking phase.

Root Imaging and Root Analysis

Root imaging was done using multispectral cameras (Svane et al. 2019b) at four dates in 2018 and three dates in 2020. Images were taken at 35-mm intervals along the minirhizotrons, from the deepest to the most shallow image, yielding almost 30,000 images per imaging date. In 2020, the total tube length was imaged, while in June 2018, the profile was screened up to 0.89 m depth, up to 1.16 m in July and in August up to 1.3 m of soil depth.

Root image analysis was done using the RootPainter software (Smith et al. 2022) trained on potato datasets from 2018.

The last imaging in 2018 was performed 1 week after harvest. At this imaging, potato roots appeared dark and thin, thus difficult to detect on the images, and the data set was accordingly excluded from further analysis. The first imaging in 2020 was performed in May, 1 month after planting and no roots were observed in soil deeper than 0.5 m. Thus, this data set was also excluded from further analysis. Imaging time points used in this study correspond to tuber initiation, tuber bulking and tuber maturation phases.

Tracer Application

In both years, nitrogen tracer ($\text{Ca}^{15}\text{NO}_3)_2$, 98% enriched) was applied to potato crops through sub-irrigation lines approximately 2.5 months after planting, when potatoes were in the tuber bulking phase.

The tracer solution was injected to all potato plots on both sides of the V-shaped bottom, similarly to the method used in Wacker et al. 2022. Isotope labelling was attempted in 2018, but it was not successful. In 2020, injection was done using two sub-irrigation driplines, corresponding to the soil interval of 1.3–1.4 m depth. Based on the assumption that the labelled area above the sub-irrigation tube was 0.3 m wide along the whole irrigation line, 85 mg $^{15}\text{N m}^{-2}$ was applied per dripline per bed. The sub-irrigation line used for the isotope injection was activated applying water in pulses for a few hours 1 day prior to injection in order to assure the soil around the irrigation line was moist before the actual injection took place. In July, volumetric water content (VWC) at the injection depth was ca 23% on the sides of the plots where minirhizotrons were placed.

Isotope injection as well as sampling and planting design was improved in 2020 (see the “[Potato Cultivars and Planting Design](#)” section) which resulted in isotope dataset of a better quality.

Sampling

The potato growth duration was approx. 130 days and all potato samples, except ^{15}N labelled samples (see below), were collected at the end of August in both

years. The shoots were removed approx. 10 days before the harvest in both years and the isotope labelled sample was collected by sampling two potato plants growing above the injection area.

In both years, each potato plot was sampled in three sections, representing the shallow, deep and isotope labelled parts of the plot (Fig. 1), while the outer most plant at each end of the row was discarded as a border plant. In 2018, the shallow part represented high water availability (sub-irrigated), due to the sub-irrigation and the deep part had lower water availability (non-sub-irrigated). In 2020, this was opposite, as the highest water availability was in the deep part, due to the deeper available soil profile (Fig. 1).

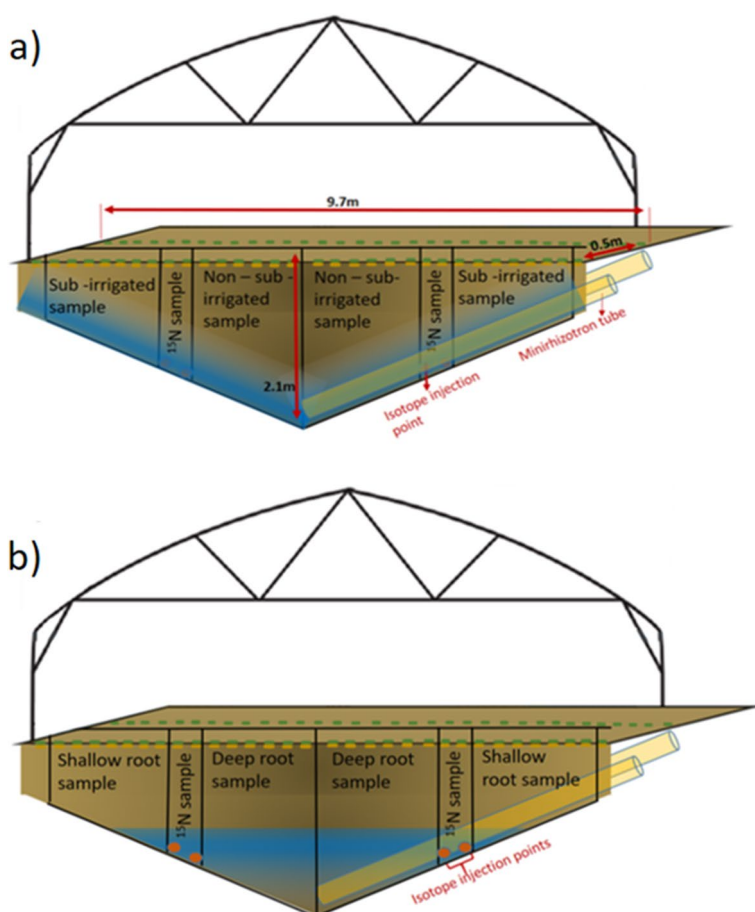


Fig. 1 Cross section of a RadiMax bed, illustrating the sampling design in (a) 2018 and (b) 2020 and cross-section of RadiMax facility. In 2018, differences in water treatment are due to irrigation system used, while in 2020, water differences are due to the soil volume in which roots are growing (more soil volume in deeper parts of soil profile). Tracer injection points are presented as red circle (one irrigation line in 2018 (1.4 m soil depth) and two irrigation lines in 2020 (1.3–1.4 m soil depth))

Right after the harvest, the total yield was weighed and representative sub-samples were taken for further analysis. Each potato tuber was cut in approx. eight pieces out of which one piece was taken as a representative part of the tuber. These subsamples were dried in the oven for 48 h at 80°, milled to 250 μm (Cyclone Mill Twister, Retch, Germany) and analyzed for ^{15}N and ^{13}C content at the Stable Isotope Facility, UC Davis, by continuous isotope ratio mass spectrometry (Sercon Ltd., Cheshire, USA). Enrichment of the tubers was expressed as δ values (Sharp 2017).

Dry matter content was measured right after subsample drying in our own laboratory.

Calculations

We applied different approaches (cut-off, cumulative root profiles and threshold) to calculate root traits, trying to get the best possible deep root estimates from the extracted root length data from the images. The traits combine information on root distribution, density and soil depth in different ways (Table 2). All calculated root traits were based on root length estimates from each image, which were square root transformed prior trait calculation. This transformation was done in order to minimize the effect of high spatial variability in root length among the images, and especially from some individual images with high root length caused by local root clustering. The total root length (TRL) and several other deep root traits adapted to potato root datasets were calculated (Table 2) (Wacker et al. 2022). Based on correlations to $\delta^{15}\text{N}$ values, we decided to select TRL calculated as sum of total root length from the whole MR tube and deep root length (DRL) calculated as sum of root length below average depth where 50% of root length are accumulated (aD50).

^{15}N uptake was calculated based on tuber total weight and N%, and was corrected for natural abundance of atmospheric ^{15}N (Junk and Svec 1958).

Statistical Analysis

Root traits associated with yield, $\delta^{15}\text{N}$ and nitrogen uptake from the tracer were selected for further analyses based on a Spearman correlation matrix.

The effect of cultivar on the calculated root traits, $\delta^{15}\text{N}$ and nitrogen uptake from the tracer was tested using a linear mixed model with bed as fixed effect and blocks within bed as nested random effects, and an exponential correlation structure to capture the spatial variation within each plot.

All models were fitted using the nlme package in R version 4.2.2 (R core Team 2022). Estimated cultivar means were determined using the emmeans package (Lenth 2022).

Results

Weather and Soil Conditions

The weather conditions in the two experimental years were very different. In 2018, temperatures were very high from mid-April combined with almost no precipitation until the end of July (Fig. 2). In 2020, most of the growing season

Table 2 Total root length and calculated deep root trait—descriptions and abbreviations

Root trait	Abbreviation	Description	Measure
Total root length	TRL	Sum of the square roots of root length from the whole MR tube	Density
Average aD50	aD50	We first calculated iD50 as depth at which 50% of root length are accumulated, individual for each tube aD50 is calculated as average iD50 values for all tubes in one imaging date	Distribution
Deep root length below average D50	DRL	Sum of root length on the individual tube below a cut-off depth defined for all tubes. The cut-off depth is calculated where 50% of all roots are observed above (aD50) at the observation date The values are calculated based on square roots of root length per image In this study we have calculated following: June 2018: aD50=63 cm soil depth July 2018: aD50=63 cm soil depth August 2018: aD50=68 cm soil depth June 2020: aD50=54 cm soil depth August 2020: aD50=73 cm soil depth	Density

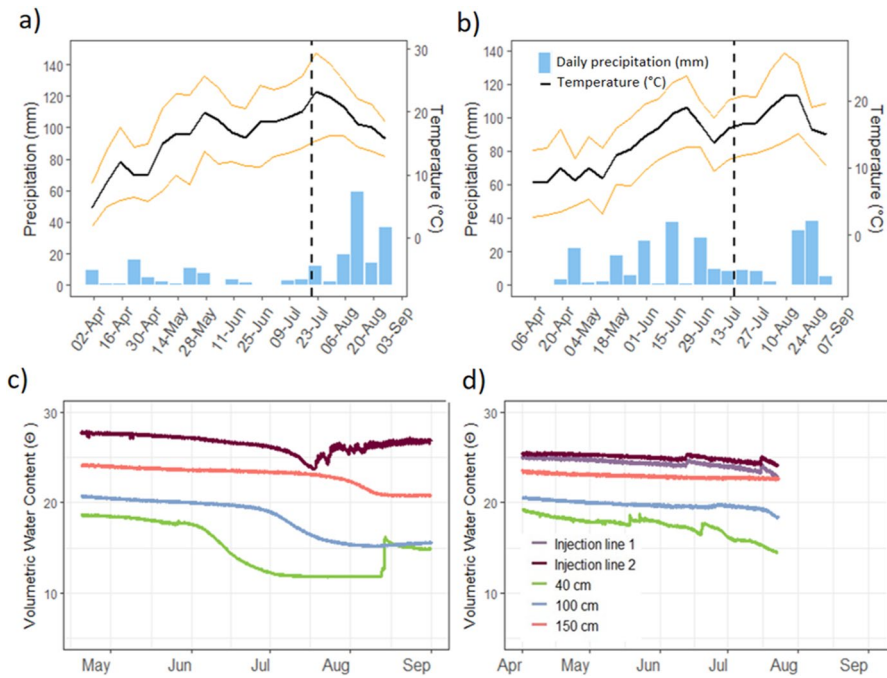


Fig. 2 Weather data and soil volumetric water content during potato growing season in 2018 (**a, c**) and in 2020 (**b, d**). Total daily precipitation (mm) and daily average temperature (°C) from April to September in both years at Taastrup meteorological station, located 600 m from experimental plots. Dashed vertical black line indicates the date of isotope injection

was rather humid and with average temperatures lower than in 2018, although precipitation in May, when potatoes were just established, was low.

The dry weather of 2018 led to high water consumption (Fig. 2) and soil water content decreased to 12% VWC at 40 cm soil depth, while in 2020, a minimum of 14.5% of VWC was reached in late July. In July, at the injection time, VWC at the injection soil depth was around 25.5% in 2018 and 23% in 2020, and the soil temperature at the tracer injection depth was on average 4 °C higher in 2018 than in 2020.

Yield and Dry Matter

Fresh yields of the low water availability samples (non-sub-irrigated in 2018 and shallow sample in 2020) in both years were similar and on average 30.5 (0.76) t ha⁻¹, while yields of samples with higher water availability (sub-irrigated in 2018 and deep sample in 2020) were lower in 2018 (35.7 (0.91) t ha⁻¹) than in 2020 (41.7 (0.75) t ha⁻¹) (Fig. 3). The yield of isotope labelled sample from 2020 was 40.0 (0.81) t ha⁻¹.

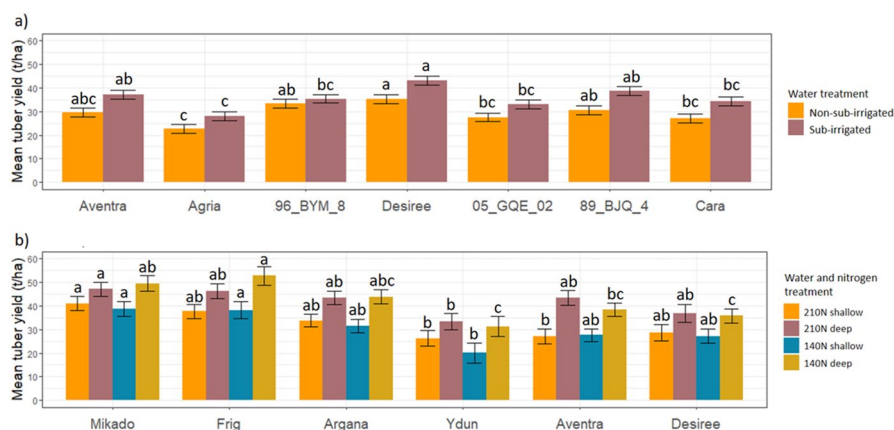


Fig. 3 Effect of water and water under nitrogen treatment on fresh tuber yield in 2018 (a) and 2020 (b). Mean tuber fresh yield and standard error bars ($n=8$) are calculated as a mean of fresh yield for each water in 2018 (a) and interaction of water and nitrogen treatment in 2020 (b). For 2020, presented is a combination of effect of water stress (shallow and deep) under high (210N) and normal (140N) nitrogen treatment. Letter of significance indicates differences between cultivars in their fresh yield under the same water condition in 2018 and under the same combination of water and nitrogen treatment in 2020. Treatments that share a letter are not significantly ($p < 0.05$) different

Cv. Desiree showed stable yields across years, nitrogen and water treatments, yielding on average 34 t ha^{-1} with only high water availability sample from 2018 yielding 43 t ha^{-1} . In 2020, the earliest cultivars (cv. Frig and cv. Mikado) had the highest yields ($\sim 40 \text{ t ha}^{-1}$).

Dry matter content of tubers was not significantly ($p > 0.05$) affected by drought treatment, but it was significantly lower in 2018 (19.1% (0.4)) than in 2020 (26.9% (0.5)) (data not shown).

The level of nitrogen fertilization (140 or 210 kg N ha^{-1}) applied in year 2020 did not affect the yield or dry matter content of potato tubers ($p > 0.05$).

Roots—Growth and Distribution

In both years, all genotypes grew substantial root length in the studied soil layers below 0.5 m with most deep roots observed in August. Substantial root length was observed below 0.7 m and roots were detected to 1.4 m soil depth (Fig. 4). In general, through all months, more roots were found in 2018 than in 2020, yet deeper roots were found in the root profile from 2020. No effect of nitrogen dose on root distribution and depth was observed. In both years, shifts in root growth during the season were observed. On average, in soil layers above 0.9 m and 0.7 m soil depth in 2018 in 2020, respectively, potato cultivars had more roots in earlier months (June and July) compared to August (Fig. 4).

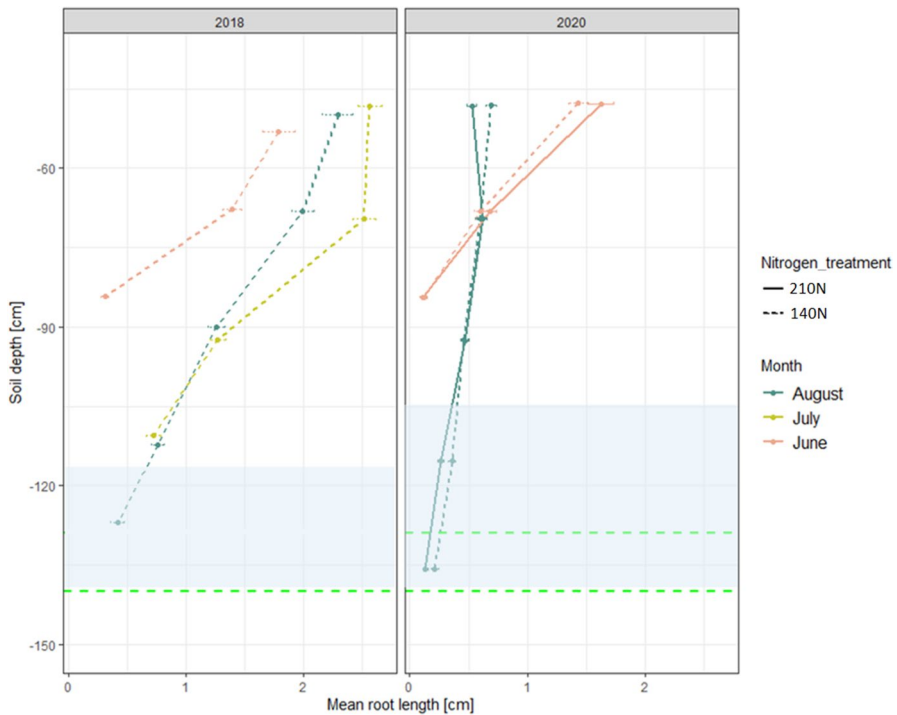


Fig. 4 Root distribution in soil profile (0.4–1.4 m) based on average root length per 3.5-cm image interval for each imaging campaign. Points represent means with standard error bars for root length in corresponding soil depth interval. Horizontal green line indicates tracer injection depth in both years (2018 = 1.4 m, 2020 = 1.3–1.4 m), while blue shaded box accounts for water movement upward (25 cm) due to the capillary rise

For most genotypes, the root length measured through minirhizotron tubes (total root length (TRL)) was highest in July for 2018 while in 2020 cultivars had higher TRL values in August when compared to June (Fig. 5a, b). In both years, significant differences among genotypes in TRL and deep root measurements (DRL) were observed among genotypes at later stages, while no significant differences were found in June (Fig. 5a, b). Cara had the highest TRL at the two later imaging dates (July and August) when compared to the other genotypes grown in 2018, and this difference was significant ($p < 0.05$) when compared to TRLs of cultivars Aventra and Agria (Fig. 5a). In August of 2020, Ydun and Desiree had significantly more roots (Fig. 5b) than cultivars Aventra, Mikado and Frig (all $p < 0.05$). In both years, cv. Aventra had rather low TRL, while cv. Desiree was intermediate. In August 2020, the two early finishing cultivars, Frig and Mikado, had significantly ($p < 0.05$) lower TRL compared to the other cultivars (Fig. 5b).

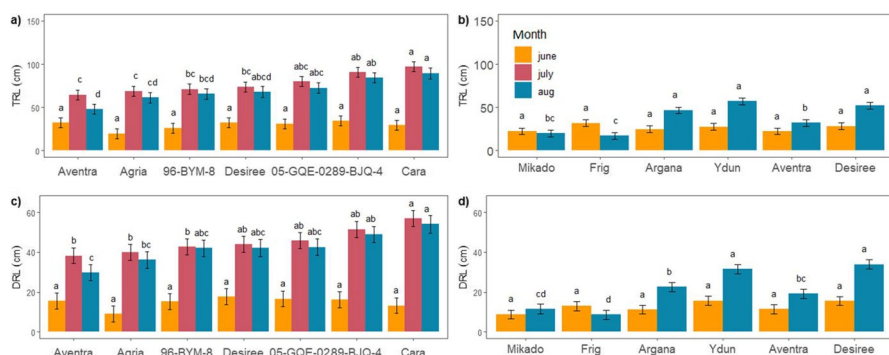


Fig. 5 Total and deep root length calculated for each potato cultivar. **a, b** Total root length (TRL) calculated as root length sum in each individual tube, grouped by potato cultivar in 2018 (**a**) and 2020 (**b**) with standard error bars. **c, d** Deep root length (DRL) calculated as sum of root length with standard error bars below soil depth defined based on average aD50 (soil depth at which 50% of root length is accumulated) values of all tubes in 2018 (**c**) and 2020 (**d**). For more specific information about cut-offs, see Table 2. Cultivars that share a letter within the same month are not significantly ($p < 0.05$) different

The deep root characteristics described by DRL followed the pattern of TRL. In both years, the deepest roots were observed in August and cultivars ranked similar as with respect to the TRL (Fig. 5c, d).

Isotope Tracers

Potato Tuber ^{15}N Enrichment

High nitrogen dose did not significantly affect total nitrogen uptake and nitrogen recovery from the tracer (Fig. 6a, b) when compared to uptake and recovery at 140 N dose. Total nitrogen uptake was on average $135.8 \text{ kg N ha}^{-1}$ and there was no significant ($p > 0.05$) difference between cultivars regardless of nitrogen fertilization treatment (Fig. 6a).

^{15}N labelled N was injected at 1.3–1.4 m depth and the calculated nitrogen recovery in the tubers was used as an indicator of uptake of nitrogen from deeper soil layers. Significant ($p < 0.05$) differences among genotypes in ^{15}N enrichment in tubers were found (Fig. 6a) on high nitrogen treatment (210N). Desiree cultivar recovered on average 41% of injected tracer at 210N dose, while cultivars cv. Frig, cv. Mikado and cv. Argana recovered between 0.3 and 7.5% (Fig. 6b).

Regardless of the nitrogen fertilization level, the cultivars cv. Ydun and cv. Desiree had significantly ($p < 0.05$) higher $\delta^{15}\text{N}$ than cv. Mikado and cv. Frig. At the normal nitrogen level, cv. Aventra also had significantly ($p < 0.05$) higher $\delta^{15}\text{N}$ than cv. Mikado and cv. Frig.

Cultivars Desiree and Aventra recovered more N from injected tracer at high N treatment when compared to normal, but this difference was only statistically significant for Desiree cultivar.

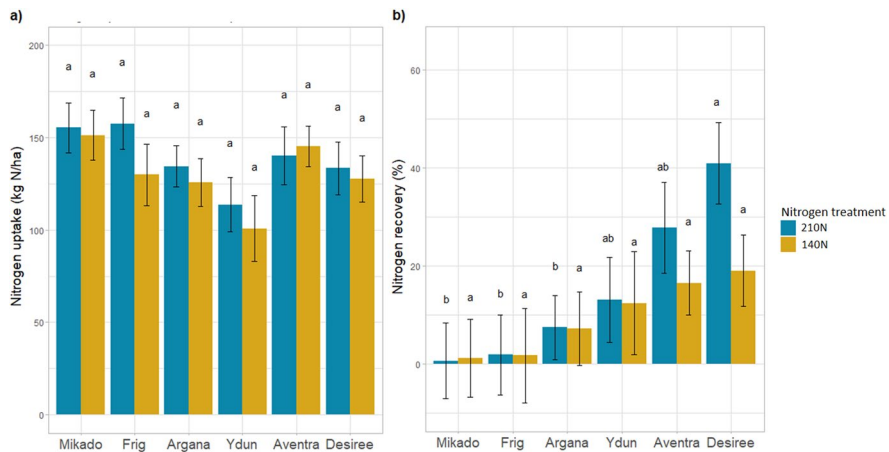


Fig. 6 Nitrogen uptake (**a**) and recovery (**b**) in isotopically labelled samples. **a** Mean total N uptake value and standard error bars measured in tubers of labelled sample at harvest in 2020 ($n_{2020}=96$). **b** Mean value of ^{15}N recovery efficiency from tracer and standard error bars in tubers of labelled sample at harvest in 2020 ($n_{2020}=96$). Calculation is presented as a ratio of mean N recovery value based on N (at%) values and dry yield, corrected for ^{15}N natural abundance in soil (0.3666) and total amount of injected isotope. Nitrogen treatment stands for nitrogen high dose (210 kg ha^{-1}) applied in July (tuber bulking phase). Treatments that share a letter are not significantly ($p < 0.05$) different

^{13}C in Potato Tubers

Based on $\delta^{13}\text{C}$ discrimination measured in water limited, less limited, and ^{15}N labelled tubers, mainly water limited samples from 2018 (non-sub-irrigated) showed higher $\delta^{13}\text{C}$ values of -21.7 (0.1) (Table 3). All other samples from both years showed lower $\delta^{13}\text{C}$ values in the range from -24 to -26 , thus indicating less water limitation in the tubers.

In 2018, cv. Desiree had the most negative $\delta^{13}\text{C}$ values in both irrigated and non-irrigated sample (-25 vs -22) (Fig. 7a). Cultivar Cara showed significantly (<0.05) higher $\delta^{13}\text{C}$ values, with -24 and -21 in sub-irrigated and non-sub-irrigated sample, respectively, when compared to cv. Desiree. All other cultivars in 2018 were not significantly different from each other nor from cv. Desiree and cv. Cara. In 2020, regardless of water treatment, cv. Ydun showed significantly (<0.05) higher $\delta^{13}\text{C}$ values (-25) than the other cultivars, while cv. Mikado showed the lowest $\delta^{13}\text{C}$ values (-27) and cv. Desiree showed intermediate results (-26) (Fig. 7b).

Table 3 Carbon discrimination in tubers. Mean values of $\delta^{13}\text{C}$ measured in non-irrigated and irrigated samples in 2018 and in dry and wet samples in 2020. Values given in the brackets indicate standard error

Sample	(n)	$\delta^{13}\text{C}$ mean	Year
Non-sub-irrigated	56	-21.7 (0.1)	2018
Sub-irrigated	56	-24.1 (0.1)	2018
Shallow root	96	-25.7 (0.1)	2020
Deep root	96	-25.6 (0.1)	2020

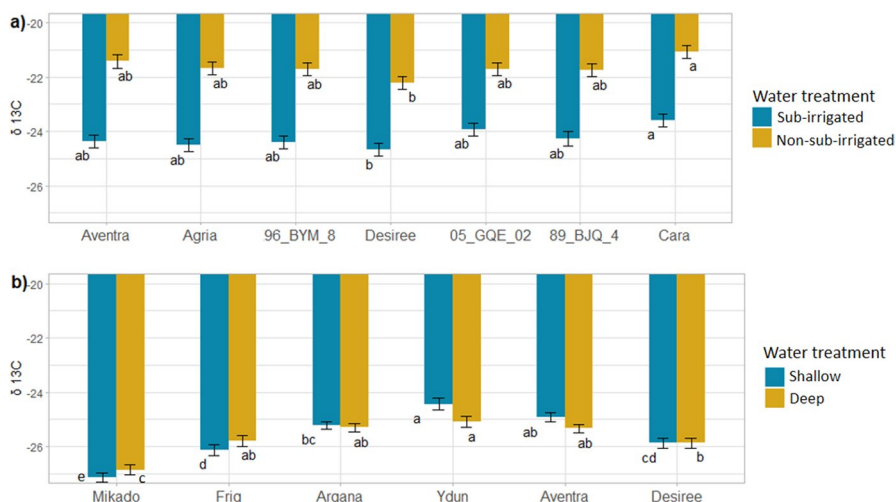


Fig. 7 Genotypic difference in ^{13}C discrimination of tubers. **a** Irrigated and non-irrigated samples in 2018 ($n_{2018}=56$). **b** Deep and shallow root samples in 2020 ($n_{2020}=96$). Letters of significance indicate genotype differences in their $\delta^{13}C$ discrimination within water treatment

Deep Root Role in Nitrogen Uptake

Deep root nitrogen uptake was evaluated through the correlation between the calculated deep root traits and $\delta^{15}N$ measured in the tubers. In general, a stronger correlation was observed with deep root data from August, compared to root traits from earlier imaging. The strongest correlation ($R=0.89$) between root and $\delta^{15}N$ measurements was observed for DRL (Fig. 8b). Moreover, high positive correlations ($R=0.83$) for the total root length (TRL) were also established (Fig. 8a).

Discussion

The RadiMax Concept for Root Quantification

The potato experiments demonstrated that the RadiMax concept (MR + water treatments + isotope tracers) has a great potential to quantify the deep rooting of a crop and to follow root development during the growing season. The use of RootPainter software with its graphical user interface and corrective training ability was of great importance for potato root quantification, particularly its efficiency which allows us to analyze large number of images. The high sensitivity of RootPainter to segment fine root structures enabled us to detect potato roots to 1.4 m soil depth in later months and allowed us to estimate root length on approx. 90,000 images each year.

Use of RadiMax in 2 years of potato experiments gave a general insight into the development of the potato root system. By observing collected images from 2018 and 2020, we found that potato roots below 0.5 m soil depth appeared to be scarce

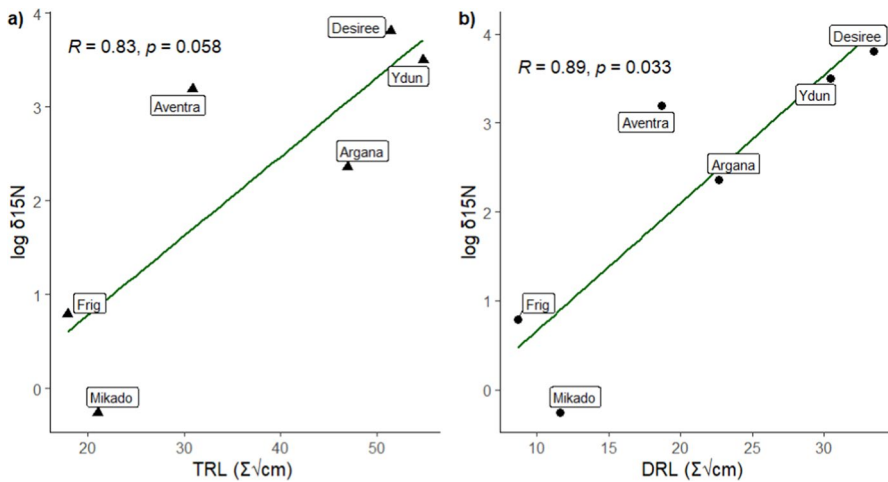


Fig. 8 Correlation of total root length (TRL) and calculated deep root characteristics (DRL) from August imaging campaign and $\log \delta^{15}\text{N}$ values (**a**, **b**) measured in tubers at harvest of 2020. For more details of calculated deep root characteristics, see Table 2. All root characteristics are calculated as square root values of root length measured at individual image. The green regression line shows the best fitting linear regression line. Each point is calculated as average root measurement from two MR tubes for one potato row, and average isotope measurement for two samples in the same potato row

and thin and that they appeared dark in colour, in contrast to bright and thick roots of wheat and grasses (Fig. 9). They tended to grow preferentially in existing soil channels and pores from previous crops, following and often hidden behind old roots, thus complicating their quantification.

As RadiMax allow repeated root imaging of crops grown to full maturity, changes in root growth patterns over the season could be observed. The greatest average root length corresponded to the period of intense above-ground growth (June and July), while downward growth continued during the whole period. The observed reduction of average root length from June/July to August above 0.9 m and 0.7 m soil depth in 2018 and in 2020, respectively, was most probably caused by a combination of factors and it has been previously reported for other crops, e.g., chicory and winter wheat (Chen et al. 2021; Odone et al. 2023; Wacker et al. 2022). Higher average root lengths were measured in 2018 than in 2020, but deeper roots were observed in 2020, and this trend in root growth can be related to the water limitation

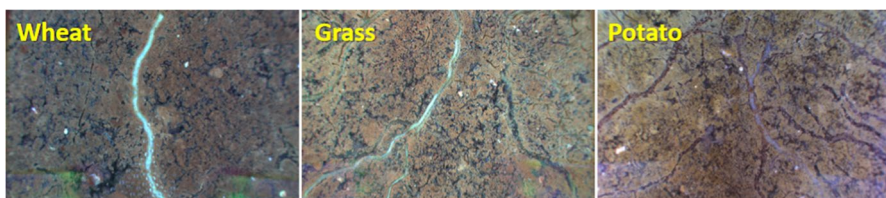


Fig. 9 Wheat, grass and potato roots imaged at RadiMax platform, showing different pattern and colours

caused stress. Several studies (Boguszewska-Mańkowska et al. 2020; Lahlou and Ledent 2005; Zarzyńska et al. 2017) indicated opposing effects of stress caused by soil water limitation on the development of deep roots systems. This relationship is not simple to identify and some discrepancy in results have been observed in previous literature. The study of Boguszewska-Mańkowska et al. (2020) reported that in response to water stress, drought tolerant potato cultivars can develop more elongated roots than sensitive varieties. However, the same study also reported that the drought-resistant cultivar (Gwiazda) previously grown in field and pot trials had relatively small root mass. Lahlou and Ledent (2005) and Zarzyńska et al. (2017) showed that cultivars with a greater root length and weight will have smaller decline in tuber yield under drought condition than those with smaller root length and weight. Despite the inconsistency, the results show that crops with larger and deeper root systems will have higher drought resistance, especially when having in mind that those studies (Lahlou and Ledent 2005; Zarzyńska et al. 2017) showed that in favourable environments without water shortage, no significant correlation between root system length and weight and yield had been established.

Deep Roots in Potatoes

The potato cultivars studied differ in their root characteristics, with respect to total root length (TRL) and deep root length (DRL). Wishart et al. (2013) conducted field trials including 28 potato cultivars grown to full maturity and sampled roots with large and small forks collecting and classifying (stolon and basal) all visible roots, but soil depth from which roots were collected was not recorded. This study established variability in rooting traits (root length and proportion of different root class, root number and root weight) between commercially available potato cultivars belonging to two distinct *Solanum* groups (Tuberosum and Phureja) and also among Tuberosum varieties. Zarzyńska et al. (2017) studied root differences of five potato cultivars in five soil depth intervals (every 0.2 m down to 1 m soil depth) at the time of potato flowering and found that cultivars significantly differed in their root length, root area, diameter and total dry weight of the roots. Later in the growing season, we measured great difference in TRL and DRL between our tested varieties, which seemed to be related to different phenology. In our study, cultivars having greatest total and deep root length were those with the long growth period, such as Cara (Stalham and Allen 2001). Additionally, cultivars Mikado and Frig that were developing fast and senescing approx. 3 weeks earlier than other cultivars were also those with the smallest root systems, as their TRL and DRL were significantly (<0.05) lower than of the other cultivars. Under optimal growing conditions with enough water and nutrient supply, late potato cultivars will grow their roots for longer time achieving higher root length, and have more time to penetrate to deeper soil layers than early maturing genotypes. When compared to shallow rooted cultivars, deep rooted cultivars are potentially less stressed in the periods between irrigation as they have access to the deep soil water later in the season and are able to uptake more of sub-soil nitrogen. When growing conditions are not optimal and potato plants are exposed to water limitation-induced stress, some cultivars show traits which help

them escape drought through early maturation, or to adapt to lower water availability. A good example of this is the cv. Desiree. Studies by Deblonde et al. (1999) and Jefferies and Mackerron (1987) showed that this cultivar was less affected by drought than other studied varieties. The main reason in this case was that under early drought stress, Desiree has weak stomatal control and maintained its photosynthetic rate and it showed capacity to preserve canopy growth, thus remain their root growth and yield tubers with lower dry matter content.

Deep Root Function—Nitrogen and Water Uptake

Despite the widespread belief that deep roots hold little significance due to their relatively small contribution to the total root biomass, previous research (Harper and Tibbett 2013; Maeght et al. 2013) highlighted their crucial role, particularly in water and nitrogen uptake and redistribution. While the upper soil layers are adequately rooted for optimal water and nitrogen uptake, the plants will depend on deeper soil layers for continued water uptake when the topsoil is dry. Commercial potato production relies on irrigation strategies which often involves high intensity of water applied to the top soil in a short time, leading to a significant water loss once water is percolated to the deeper soil layers. Deep rooted cultivars will therefore be more efficient in the use of irrigation water, increasing their resilience to water shortage between irrigation events. Additionally, deep roots play a vital role in capturing nitrogen from deeper soil layers, preventing its loss into the environment. Therefore, the stability of yields and advancements in water and nitrogen uptake hinge on enhancing rooting in the less efficiently rooted deep soil layers, a trait which has been shown to vary among genotypes (Puértolas et al. 2014).

Genotypic Differences in Total Nitrogen Uptake and ^{15}N Recovery

Genetic diversity for nitrogen uptake in potato has been previously reported (Ospina et al. 2014) and it can be partially explained by maturity type, indicating that later cultivars have higher nitrogen uptake and efficiency when compared to early maturing types. The cultivars used in this study belong to commercially classified early maturity group and have been selected for growing in modern, high input systems with lots of nitrogen fertilizer and irrigation. Given that both nitrogen doses applied in this study were high, the difference in nitrogen uptake among cultivars was not big enough to be statistically significant. Measuring nitrogen uptake in potatoes is determined by a number of both shoot and root factors, and most of its uptake is occurring in the upper soil layers, with smaller contribution of nitrogen from the subsoil. In the top soil, with high root density, root competition for nitrogen uptake is high, but in the subsoil, nitrogen uptake may be limited by rooting density and at the soil depth below 1 m, nitrate uptake can be depending on presence or absence of roots at this soil depth. Therefore, it is obvious that deep roots are important for nitrogen uptake, despite their low length/density. The effort of this study is increasing subsoil N uptake, and for measuring the cultivar differences in deep nitrogen uptake, we use ^{15}N measurements.

Some studies (Villordon et al. 2014; Wishart et al. 2013) indicate that deeper basal roots are mainly responsible for supplying plant with water and providing it anchorage, while later in the growing season, stolon roots capture nutrients and promote tuber growth. In this study, we did not classify the roots, but we documented that cultivars with larger DRL were able to recover up to 40% of ^{15}N tracer injected at 1.35 m soil depth, while for cultivars with low DRL, the uptake was as low as 0.7%. The roots observed at 1.35 m soil depth in this study are most probably basal roots with longer time to grow and with great plasticity in response to water stress, thus growing and spreading deeper in the soil.

The ^{15}N recovery in tubers varied among cultivars, and it was the lowest (0.7%) for cultivars Mikado and Frig, which was expected as these cultivars had lowest DRL and thus limited or no access to the injected isotope. The most deep rooted cultivars Desiree and Aventra had the highest ^{15}N recovery. The strong positive correlation ($R=0.8$) between root measurements (TRL and DRL) and $\delta^{15}\text{N}$ in tubers clearly showed that variation in deep root growth was important for variation in deep nitrogen uptake by the cultivars. It also shows that uptake of ^{15}N tracer applied at 1.3–1.4 m soil depth was a valid indirect measure of deep root growth through root deep root function. A substantial amount of literature has documented effects of deep roots for nitrogen uptake in cereals, vegetables and oil crops, but results on deep rooting in potato are scarce. Our findings are in accordance with those of studies previously done at RadiMax (Odone et al. 2023; Wacker et al. 2022) which established positive correlations between deep roots and nitrogen uptake in winter wheat varieties.

The high correlation between our deep root measurement (DRL) and uptake of deep placed ^{15}N tracer indicates that DRL is an appropriate measure of the role of deep roots in nitrogen uptake. The importance of root-based breeding for increased water and nutrient uptake in limiting environments is growing and thus the need for identification of useful root measurement methods. Again, substantial work has been done in cereals and maize (Changdar et al. 2023; Lynch 2013; Thorup-Kristensen and Kirkegaard 2016; Wacker et al. 2022) but a lot less is known in the field of potatoes and other root and tuber crops. Khan et al. (2016) stated that root traits such as length, number, spread and length of lateral roots are very responsive to environmental changes and increase nutrient efficiency in potatoes. While our study show deep rooting to be of critical importance for deep nitrogen uptake, more studies focusing on root architecture and how to connect it to the root function and soil transport mechanisms are needed to guide breeders towards developing potato cultivars with high water and nutrient use efficiency.

Deep Roots and $\delta^{13}\text{C}$

Water availability strongly affected fresh potato yield in both years. Interestingly, cultivars with small and short root system (Mikado and Frig), as well as water stress-resistant cv. Desiree (Deblonde et al. 1999; Jefferies and Mackerron 1987), showed strong ^{13}C discrimination in the tubers while deep rooted cultivars (Cara and Ydun) failed to do so. Overall, a weak but positive correlation of TRL and DRL to $\delta^{13}\text{C}$

values, regardless of cultivar, year and water treatment, was observed in this study, although opposite correlations have been shown in other studies (Odone et al. 2023; Wacker et al. 2022). We also observed that the cultivars with the least rooting depth and strongest ^{13}C discrimination were the earliest varieties, clearly showing that phenology plays an important role in potatoes exposure to soil water stress. One of the most well-known traits conferring drought tolerance is early maturation leading to drought avoidance. Early maturation gives less time for growth, limiting anything from deep root development to high yield potential, but at the same time, it may allow a cultivar to escape dry conditions later in the season. While the ^{15}N results showed that later maturing genotypes with deep root growth did increase subsoil exploration, presumably also for water, then the early maturing cultivars with less rooting depth avoided drought by finishing growth early, and thereby showing stronger ^{13}C discrimination. While “drought avoidance” can be a good strategy against water stress in environments with strong summer droughts, it is less clear how valuable this strategy would be for northern Europe. This region has significant summer rainfall, and yield potential would be compromised as early maturing cultivars will fail to use the summer rainfall for yield formation.

The strong effect of phenology in our study could be a consequence of sampling time. The value of $\delta^{13}\text{C}$ in a plant organ represents an integrated measurement of conditions during the time of formation and growth of that particular organ. Thus, $\delta^{13}\text{C}$ values of tubers sampled at maturity are representing accumulated ^{13}C discrimination during the tuber bulking phase. Approximately 3 weeks prior to the harvest (sampling time), we observed loss of canopy for early cultivars indicating they already stopped their photosynthesis and tuber formation, while the later ones continued their tuber bulking into drier summer periods until the harvest. Thus, sampling the young potato leaves right before flowering, which is the time before significant phenology effect appears, might be a better indicator of the effect of deep root activity on plant water status under water stress. The field study of Wishart et al. (2013) reported strong correlation between root weight and carbon discrimination in potato leaves, showing that the larger the root weight is, the greater the discrimination against heavy isotope is measured. However, this early measurement will not represent higher drought stress which normally happens later in the season.

Another possibility for measuring deep root function in water uptake could be the use of ^2H tracer injected at the certain soil depth. Previous studies on spring (Ober et al. 2014) and winter wheat (Odone et al. 2023) indicated that varieties with deeper roots were more drought tolerant. The study of Odone et al. (2023) showed ^2H labelling is a valuable indicator of deep soil water uptake increasing the potential for overcoming water stress.

Based on the results obtained here and supported with the above-mentioned literature, it is obvious that $\delta^{13}\text{C}$ values of tubers have a complex relationship to deep root water uptake and water stress status of potato crops in general. There are many factors involved in making a cultivar drought resilient, not least the differences in phenology. Measurements of ^{13}C discrimination in tubers alone are effectively representing deep root water uptake and root trait-related varietal water stress resilience.

Conclusion

We have found that RadiMax root phenotyping platform, coupled with AI-based software (RootPainter), effectively captures and quantifies the deep root growth of potato.

The calculated root traits (TRL and DRL) exhibited variability among potato varieties, and also their capacity for deep nitrogen uptake. Maturity group strongly influenced these traits, and especially the water stress levels through drought escape.

Under the umbrella of climate change and increased food demand, future potato production relies a lot on robust potato genotypes that can grow well in both dry and wet conditions. With that in mind, the desired genotype should have a combination of traits to ensure water stress tolerance and high nutrient uptake. Results of this study showed that deep roots are beneficial for increased deep nitrogen uptake, but that phenology of potatoes is often overpowering the direct effects of root traits on drought stress. The results highlight a significant genetic diversity in deep nitrogen uptake among the studied varieties within the same maturity group presenting opportunities for breeding improvements. However, among the cultivars included in this study, deep rooted genotypes did not exhibit higher water stress resilience. It is still likely that if cultivars with the same development time (phenology) are compared, especially late cultivars, the deep rooting giving access to deep soil water is important, as has been shown with other crops.

Future research is imperative to comprehend the water uptake mechanisms of potato crops under soil water stress. The method should be refined to elucidate varietal differences in root traits, and its contribution to improved soil water acquisition.

The major social and economic impact of the study is to contribute to sustainable intensification of potato production based on stable yields without input increase and under changing climates. This can be achieved by breeding for deep-rooted potato cultivars with better use of irrigation water, thus better performance between irrigation events and with potential for reducing nitrogen losses through leaching.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11540-024-09802-4>.

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Data Availability The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

Competing Interests The authors declare no competing interests.

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